

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 1 of 12

(1) Research Program/Project/Study Title: GRACE: Grit, Academic Resilience, and Emotional Intelligence of Education Students	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Pilar
Province:	Capiz
Region:	VI

(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity Education/ Social Sciences	(8) Discipline/Field of Specialization English Mathematics
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	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 2 of 12

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

It is important in tertiary education especially in the education department be gritty, and be academically resilient and emotionally intelligent to be successful in the teaching world. Hence, this study was conducted to find out the level the level of grit, academic resilience, and emotional intelligence of the education students as a whole and when grouped according to their age, sex, civil status, number of siblings, parents' occupation, parents' highest educational attainment, and estimated monthly family income. This also aimed to determine if there was a significant relationship between grit and emotional intelligence, grit and academic resilience, and emotional intelligence and academic resilience. Using the Grit Scale, SSEIT, and Academic Resilience Scale, the data were gathered among the education students of Capiz State University- Pilar. The results showed that the respondents were gritty, highly academic resilient, with above average emotional intelligence. It further revealed there was a weak significant correlations among grit and emotional intelligence, and grit and academic resilience; and academic resilience and emotional intelligence had strong significant correlation. It is recommended to have a holistic approach and to tailor these strategies to the socio- economic status of the students further develop their grit, emotional intelligence and academic resilience which were important characteristics in the teaching world.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Student motivation, psychological resiliency, academic performance, socio- economic status

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

College education is considered as the make-or-break point in a person's life, especially here in the Philippines as education is valued highly and that it is easier to land a job when you have a bachelor's degree. However, college can be tough, especially in education students as they will serve as future teachers and role models of the youth. They must uphold the highest morale, be mentally alert, must have excellent scholastic standing, and be emotionally stable as they are trained to be strong and ready before facing the real-world of teaching. As soon to be teachers, they must not only be able to pass exams, make excellent lesson plans and instructional materials, but they also need grit and emotional intelligence and how these impact their academic resilience.

Grit is passion and perseverance for long term goals which drives one to continue and pursue the goals even if there are setbacks or slow progression (Duckworth, A., 2024).

According to Deepak, A. & Krishnamoorthy, K. (2016), one of the important soft skills in tertiary is the emotional intelligence, and ability to meanings behind emotions, relationships, and problems based on it.

With the hurdles that education students face every day, it is important that they are resilient especially academically. According to Ye, W., Strietholt, R. & Blömeke, S. (2021) academic resilience refers to students' capacity to perform highly despite a disadvantaged background.

Entering tertiary education is no easy task as students face several challenges before achieving the diploma. One of the keys in unlocking this long-term goal is for the students to have grit and not stopping until it is fully accomplished. Gritty students compared to those who are not will most likely be able to complete their degrees despite the hardships as they tend to persevere more and sees the obstacles as only setbacks and not dead ends. Also, students with strong emotional intelligence will be more flexible and adaptable to change, thus they have healthy well- being with healthy relationships and connections especially to their students. With this, they must also be academically resilient to be able succeed in the face of adversity and preserve more to become successful and achieve their long-term goals.

Hence, this study was conducted to determine the level of grit and emotional intelligence of the Bachelor of Elementary Education (BEED) students of Capiz State University (CAPSU) Pilar Satellite College. The result of this study was beneficial to the Teacher Education Department as it will serve as basis on where the department the institution would address the challenges that the students are facing. This also assessed the preparedness of the students in entering the world of teaching and to improve the quality of the education graduates. Furthermore, this will contribute to innovations and curriculum improvement in order, not only to achieve high percentage on licensure exam for teachers, but also to produce quality and globally competitive graduates which is line with Sustainable Developmental Goals 3 (Good Health and Well- being) and 5 (Quality Education).

(11) Materials and Methods or Methodology

Research Design

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 3 of 12

This research utilized the cross-sectional research design, where the researcher did not control or manipulate any of the variables, but only observed and measured them. A quantitative cross-sectional research design involves analyzing data from a population at a single point in time to measure prevalence, describe health characteristics, and establish preliminary evidence for future studies (Dibya & Soumyajit, 2023; Puspa, 2022). This was an appropriate method to use when the research aims to identify characteristics, frequencies, trends, and categories (McCombes, 2019). The study also utilized the causal-comparative method to draw relationships, differences, and conclusions and variables involved in this study.

Respondents

A priori power analysis was conducted using G*Power software to determine the minimum required sample size. The analysis considered a desired significance level of .05, an anticipated effect size of .15, and a desired statistical power of 80%. Based on this analysis, a minimum sample size of 84 college students from Capiz State University was required. Stratified random sampling was then used to select participants from both working and non-working student populations of Capiz State University Pilar Satellite College, enrolled during the second semester of 2023-2024. A total of 100 respondents were invited to secure that a minimum number was reached. After data cleaning only 84 responses were considered valid, which satisfied the number of respondents for this study. Table 1 shows the profile of the respondents. Most of the respondents were 20-21 years old (39.3%), all of them happened to be female (100.00%), single (97.6%), with mothers having no occupation (54.8%), fathers with other jobs not mentioned in this study (65.5%), and belonged to monthly family income which as P5,999 and below (84.5%).

Table 1

Profile of the Respondents

Socio-Demographic Profile		Frequency	Percentage
Age	17-19 years old	32	38.1
	20-21 years old	33	39.3
	22-23 years old	15	17.9
	24 years old and above	4	4.8
Sex			
	Female	84	100.0
Civil Status			
	Single	82	97.6
	Married	2	2.4
Mother's Occupation			
	Teacher/Instructor	2	2.4
	Business Woman/Entrepreneur	1	1.2
	Barangay Official	3	3.6
	Government Employee	3	3.6
	None	46	54.8
	Others	29	34.5
Father's Occupation			
	Businessman/Entrepreneur	1	1.2
	Barangay Official	1	1.2
	Government Employee	4	4.8
	None	23	27.4
	Others	55	65.5
	Businessman/Entrepreneur	1	1.2
Estimated Monthly Family Income			
	P5,999 and below	71	84.5
	P6,000-P10,999	8	9.5
	P11,000-15,999	1	1.2
	P16,000-P20,999	2	2.4
	P21,000 and above	2	2.4
Total		84	100.0

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 4 of 12

Time and Place of the Study

This study was conducted at CAPSU Pilar during the 2nd semester, A. Y. 2023- 2024.

Research Instrument

The socio- demographic profile of the respondents was collected for in the first part of the questionnaire. Information about their age, sex, civil status, number of siblings, parents' occupation, parents' highest educational attainment, and estimated monthly family income will be collected in the first part of the questionnaire.

The instruments used to gather grit, emotional intelligence, and academic resilience were content validated by psychology, statistics, and English experts considering the item level content validity index (I-CVI) by Polit et al (2007) and revisions suggested by the experts to address cultural difference of these adapted international scales.

For the second part, the 12-Item Grit Scale of Duckworth, A. et. al., (2007) was utilized to collect data on the grit of the respondents. The Grit Scale had the internal consistency of 0.85 (Cronbach's Alpha).

Third part of the questionnaire was for the data on the emotional intelligence of the respondents which were collected through Schutte Self-report Emotional Intelligence Test (SSEIT) by Schutte, et. al. (1998), with Cronbach's Alpha of 0.90 which showed good internal consistence and reliable.

The last part of the instrument was for the data on the academic resilience of the respondents where data would be collected through Academic Resilience Scale (ARS-30) of Cassidy, S. (2016). ARS- 30 has a Cronbach's Alpha on 0.90 which indicated high internal consistency reliability for the global scale.

Data Gathering Procedure

The researcher constructed a permission letter to be approved by the Satellite College Director and the Campus Research Chair to allow the conduct of the study. Once approved, the researchers personally administered the questionnaires to the target respondents. After the questionnaires were answered, those were collected, and the data were tallied and analyzed statistically using SPSS software.

Ethical Considerations

The respondents can withdraw their participation anytime in the duration of the collection of data as it is purely voluntary and does not involve any monetary compensation. The data collected and personal information of the respondents will be strictly confidential and will be used for the sole purpose of this research.

Statistical Data Analysis

The data gathered from the study was analyzed using the Statistical Package for Social Sciences (SPSS) software.

Frequency Count and Percentage. Frequency and percentage was used to determine the number of respondents in every section.

Mean. This was employed in determining the level grit, emotional intelligence, and academic resilience of the respondents.

Standard Deviation. This test was to determine the homogeneity as well as heterogeneity of the scores obtained and of the variables.

The data were subjected to normality test using Kolmogorov-Smirnov test and Shapiro Wilk Test. Inspecting the normality of variables, the variables turned out to be coming from normal distribution ($p > .05$). Thus, parametric tests were used for inferential questions testing the difference and relationships among data.

t-test. This test is used to determine if there is significant difference on constructs when respondents are compared by sex (male and female groups).

F-test. This test was used in comparing the constructs when respondents are grouped according to their profile, specifically, age, mother's occupation, father's occupation, and estimated monthly family income.

Pearson r. To determine the relationship of the variables, Pearson r was used.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Grit of Students

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 5 of 12

The students were found to be gritty ($M=3.15$, $SD=0.35$) as shown in Table 2.

The students who aged 24 years old and above ($M=3.48$, $SD=0.41$) were very gritty. Younger students, those who were 17-19 years old ($M=3.22$, $SD=0.28$), 20-21 years old ($M=3.17$, $SD=0.39$), and 22-23 years old ($M=2.89$, $SD=0.24$) were gritty. Furthermore, F-test revealed that there is a significant difference, $F(3,80) = 4.837$, $p = .004$, on grit of those who belonged to 22-23 years old which is lower compared to other age groups (17-19 years old, 20-21 years old, and 24 years old and above). This result suggests that age might be a factor influencing the grit possessed by education students. This is in line with the study of Kannangara, et.al. (2018) which stated that there is a positive correlation between grit and age. It also stated that the older an individual is, the higher is their level of grit. It can be implied grit increases with age; that the older the person is, the grittier he or she becomes because of the number of challenges that were surpassed.

The students who were married ($M=4.08$, $SD=0.00$) were very gritty, while those who were single ($M=3.13$, $SD=0.32$) were gritty. Furthermore, F-test revealed that there is a significant difference, $t=4.141$, $p = .000$, on grit between single and married education students. This result suggests that civil status might be a factor influencing the grit possessed by education students. This is in the opposite of the study of Roxas, E. D. (n.d.), which stated that grit and civil status had no significant difference. However, in this study, it can be implied that married individuals are very gritty because they had to manage different roles, especially with their relationships with their spouses and children, and making sure that their homes are peaceful and all the needs of the family members are taken care of.

When grouped according to mother's occupation, the students whose mothers were barangay officials ($M=3.44$, $SD=0.57$) and teachers/instructors ($M=3.42$, $SD=0.83$) were very gritty. The students with mothers had other occupation (not mentioned here) ($M=3.17$, $SD=0.35$), no job ($M=3.14$, $SD=0.30$), and those who were government employees were gritty. The students with business woman/entrepreneur mothers ($M=2.17$) were slightly gritty. The difference on grit when students are grouped according to their profile were further determined using ANOVA or F-test. The results showed that there exists a difference, $F(5, 78) = 2.572$, $p = .033$, on grit of students when they are grouped according to their mothers' occupation. Thus, mother's occupation can affect the grit of education students. This result is in line with the study of Simones Landry, M. (2018) which stated that grit and sponsorship for females drives them to lead and to aim for more leadership positions. It can be implied from this that women who had greater responsibility like barangay officials and teachers were grittier because they had bigger decisions to make compared to occupations mentioned in this study. It can also be implied that they were grittier because they had to stick longer in their responsibilities to achieve positive results.

The students whose fathers were government employee ($M=3.25$, $SD=0.45$), had other occupation (not mentioned here in the study) ($M=3.17$, $SD=0.35$), no job ($M=3.16$, $SD=0.39$), and barangay officials ($M=2.92$) were gritty. The students with businessman/entrepreneur father ($M=2.17$) were slightly gritty. F-test revealed that there is no significant difference, $F(4, 79) = 2.326$, $p = .064$, on grit when education students were grouped according to their father's occupation. These findings suggest that father's occupation is not a significant predictor of grit of education students. This result is in line with the study of Arco-Tirado et al. (2019) which stated grit and entrepreneurship had a simple negative correlation with each other.

Considering the estimated monthly family income, the students whose family earned P11,000 – P15,999 ($M=3.92$), and P21, 000 and above ($M=3.42$, $SD=0.83$) were very gritty. The rest of the students who came from a family whose estimated monthly income were P6, 000 – P10,000 ($M=3.35$, $SD=0.34$), P5, 999 and below ($M=3.13$, $SD=0.32$), and P16, 000 – P20, 999 ($M=2.67$, $SD=0.00$) were gritty. There is a significant difference, $F(4, 79) = 3.579$, $p = .010$, on grit of students when grouped according to their monthly family income. Thus monthly income of family affects grit of education students. This is in line with the study of Arco-Tirado, J. L., Bojica, A., Fernández-Martín, F., & Hoyle, R. H. (2019) which stated that more socioeconomically privileged individuals had stronger beliefs and can sustain their perseverance toward their goals. From this, it can be implied that those who came from families with higher income may had access to resources where they were comfortable to pursue their long term goals and were not afraid to fail and try again as they had more stable finances they can fall back into. It can be also be implied that they had strong support system outside school and were able to enjoy extracurricular activities like enrolling in other classes like music and sports where they developed their confidence and resiliency compared to those low- income households who lacked these opportunities.

Table 2

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 6 of 12

Level of grit of students when grouped according to their profile

Profile	M	SD	Academic Resilience Interpretation	F-test/ t-test	p-value
Age					
17-19 years old	3.22	0.28	Gritty ^b	4.837	.004**
20-21 years old	3.17	0.39	Gritty ^b		
22-23 years old	2.89	0.24	Gritty ^a		
24 years old and above	3.48	0.41	Very Gritty ^b		
Civil Status					
Single	3.13	0.32	Gritty	4.141	.000**
Married	4.08	0.00	Very Gritty		
Mother's Occupation					
Teacher/Instructor	3.42	0.83	Very Gritty	2.572	.033*
Business Woman/ Entrepreneur	2.17	.	Slightly Gritty		
Barangay Official	3.44	0.57	Very Gritty		
Government Employee	3.00	0.00	Gritty		
None	3.14	0.30	Gritty		
Others	3.17	0.35	Gritty		
Father's Occupation					
Businessman/Entrepreneur	2.17	.	Slightly Gritty	2.326	.064 ^{ns}
Barangay Official	2.92	.	Gritty		
Government Employee	3.25	0.45	Gritty		
None	3.16	0.30	Gritty		
Others	3.17	0.35	Gritty		
Estimated Monthly Family Income					
P5,999 and below	3.13	0.32	Gritty	3.579	.010**
P6,000-P10,999	3.35	0.34	Gritty		
P11,000-15,999	3.92	.	Very Gritty		
P16,000-P20,999	2.67	0.00	Gritty		
P21,000 and above	3.42	0.83	Very Gritty		
Grand Mean	3.15	0.35	Gritty		

Scale: 4.21-5.00 Extremely Gritty; 3.41-4.20 Very Gritty; 2.61-3.40 Gritty; 1.81-2.60 Slightly Gritty; 1.00-1.80 Not at all Gritty
 **p-value is significant at 0.01 level; *p-value is significant at 0.05 level; ^{ns} p-value is non-significant at 0.05

Academic Resilience of Students when Grouped According to Profile

The academic resilience of students when grouped according to profile is shown in Table 3. Generally, the respondents were found to be academically resilient ($M=3.78$, $SD=0.36$).

Across ages, 22-23 years old ($M=3.81$, $SD=0.35$), 20-21 years old ($M=3.80$, $SD=0.44$), 17-19 years old ($M=3.79$, $SD=0.29$), and 24 years old and above ($M=3.42$, $SD=0.07$), the students were found to be highly academically resilient. F-test revealed that there is no significant difference, $F(3, 80) = 1.366$, $p=.259$, on academic resilience when education students were grouped according to age. These findings suggest that age is not a significant predictor of academic resilience of education students. This result agrees with Robbins et al. (2018) who identified that age was a significant predictor of academic resilience in female students within Higher Education, according to the study's findings on predictors of student resilience.

When grouped according to civil status, single ($M=3.78$, $SD=0.36$) and married ($M=3.72$, $SD=0.54$) students were also found to be highly academically resilient. When grouped according to civil status, t-test revealed that there is no significant difference, $t = .241$, $p=.810$, on academic resilience of education students. These findings suggest that civil status is not a significant predictor of academic resilience of education students, as supported by the study findings of Blair et al. (2024).

The students whose mothers were government employee ($M=4.13$, $SD=0.00$), even those without occupation ($M=3.86$, $SD=0.39$) and other occupation ($M=3.67$, $SD=0.26$), barangay official ($M=3.65$, $SD=0.34$), teachers/ instructors ($M=3.55$, $SD=0.35$), were found to be highly academically resilient. Students with business woman/ entrepreneur mothers were academically resilient ($M=2.90$). F-test revealed that there is a significant difference, $F(5, 78) = 3.289$, $p=.010$, on academic resilience when education students were grouped according to their mother's occupation. These findings suggest that mother's occupation is a significant predictor of academic resilience of education students. This result supports the study on the occupation of a mother influences academic resilience in education students, as shown in the study on Colombian university mothers (Alejandra, et al., 2021).

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 7 of 12

The results also revealed that students with fathers who were government employee ($M=3.93$, $SD=0.30$), who has other occupation ($M=3.78$, $SD=0.37$), or who has no occupation at all ($M=3.77$, $SD=0.32$), or worked as barangay official ($M=3.63$), were highly academically resilient. Students whose fathers were in business ($M=2.90$) were found to be academically resilient. F-test revealed that there is no significant difference, $F(4, 79) = 1.754$, $p=.146$, on academic resilience when education students were grouped according to father's occupation which agreed with Blair et al.(2024).

No matter what the estimated monthly family income of the students, may it be P11,000-15,999 ($M=3.97$), P5,999 and below ($M=3.81$, $SD=0.37$), P6,000-P10,999 ($M=3.58$, $SD=0.26$), P21,000 and above ($M=3.55$, $SD=0.35$) or P16,000-P20,999 ($M=3.40$, $SD=0.00$), students were found to be academically resilient. There is no significant difference, $F(4, 79) = 1.635$, $p=.174$, on academic resilience when education students were grouped according to estimated family income. These findings suggest that the income of the family where the students belong is not a significant predictor of academic resilience, as supported by the study of Yan and Xiaosong (2022).

This result is congruent with the study of Ye et al. (2021) which stated that the classification of resiliency did not change that much in different backgrounds. This implies that the challenges in the university of the students can be easily overcome by them. They can easily find solutions and perform well in their studies despite the demand of the courses. This also meant that academic resiliency is not exclusive for those who were more financially abundant or older, but across the demographics, thus the school may continue to foster positive and resilient environment.

Table 3

Level of Academic Resilience of Students when Grouped According to Profile

Profile	M	Academic Resilience SD	Interpretation	F-test/ t-test	p-value
Age					
17-19 years old	3.79	0.29	Highly Academically Resilient	1.366	.259 ^{ns}
20-21 years old	3.80	0.44	Highly Academically Resilient		
22-23 years old	3.81	0.35	Highly Academically Resilient		
24 years old and above	3.42	0.07	Highly Academically Resilient		
Civil Status					
Single	3.78	0.36	Highly Academically Resilient	.241	.810 ^{ns}
Married	3.72	0.54	Highly Academically Resilient		
Mother's Occupation					
Teacher/Instructor	3.55	0.35	Highly Academically Resilient	3.289	.010**
Business Woman/ Entrepreneur	2.90	.	Academically Resilient		
Barangay Official	3.65	0.34	Highly Academically Resilient		
Government Employee	4.13	0.00	Highly Academically Resilient		
None	3.86	0.39	Highly Academically Resilient		
Others	3.67	0.26	Highly Academically Resilient		
Father's Occupation					
Businessman/Entrepreneur	2.90	.	Academically Resilient	1.754	.146 ^{ns}
Barangay Official	3.63	.	Highly Academically Resilient		
Government Employee	3.93	0.30	Highly Academically Resilient		
None	3.77	0.32	Highly Academically Resilient		
Others	3.78	0.37	Highly Academically Resilient		
Estimated Monthly Family Income					
P5,999 and below	3.81	0.37	Highly Academically Resilient	1.635	.174 ^{ns}
P6,000-P10,999	3.58	0.26	Highly Academically Resilient		
P11,000-15,999	3.97	.	Highly Academically Resilient		
P16,000-P20,999	3.40	0.00	Highly Academically Resilient		
P21,000 and above	3.55	0.35	Highly Academically Resilient		
Grand Mean	3.78	0.36	Highly Academically Resilient		

Scale: 4.21-5.00 Exceptionally Academically Resilient; 3.41-4.20 Highly Academically Resilient; 2.61-3.40 Academically Resilient ; 1.81-2.60 Developing Academically Resilient; 1.00-1.80 Not Academically Resilient

**p-value is significant at 0.01 level; ^{ns} p-value is non-significant at 0.05

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 8 of 12

Emotional Intelligence of Students

Students' emotional intelligence was above average ($M=3.94$, $SD=0.42$) as shown in Table 4.

Across ages, 22-23 years old ($M=3.98$, $SD=0.45$), 24 years old and above ($M=3.97$, $SD=0.11$), 17-19 years old ($M=3.92$, $SD=0.40$), and 20-21 years old ($M=3.92$, $SD=0.45$), students have above average emotional intelligence. There is no significant difference, $F(3, 80) = .072$, $p=.975$, on emotional intelligence when education students were grouped according to age. These findings suggest that age is not a significant predictor of emotional intelligence of education students. The consistent result among the different age groups agreed with the study of Kundu, D. (2019), which stated that age does not predict emotional intelligence but develops with age. It can be implied that age did not affect a person's emotional maturity as he or she may have had varied life experiences. One may also conduct a longitudinal study to explore other age ranges and demographics.

The students who are married ($M=4.34$, $SD=0.47$) have very high emotional intelligence while those who are single ($M=3.93$, $SD=0.41$) have above-average emotional intelligence. When grouped according to civil status, t-test revealed that there is no significant difference, $t = 1.380$, $p=.171$, on emotional intelligence of education students. These findings suggest that civil status is not a significant predictor of emotional intelligence of education students. This result aligned with Khodarahimi, S. (2015) who stated that married individual had higher emotional intelligence compared to other civil status. It can be implied that married individuals had higher emotional intelligence because they had heavier responsibilities at home as wives and mothers who needed to understand their husbands and children in all aspects compared to those who were single. With this, one may further investigate the aspect of marriage and singlehood which contribute to lower or higher emotional intelligence.

Those students whose mothers are barangay officials ($M=4.25$, $SD=0.52$) have very high emotional intelligence, while the rest, mothers who have no occupation ($M=3.99$, $SD=0.43$), teacher/instructor ($M=3.91$, $SD=0.60$), other jobs ($M=3.88$, $SD=0.36$), government employee ($M=3.76$, $SD=0.05$), and businesswoman/entrepreneur ($M=2.97$), all have above average emotional intelligence. F-test revealed that there is a significant difference, $F(5, 78) = 1.903$, $p=.130$, on academic resilience when education students were grouped according to their mother's occupation. These findings suggest that mother's occupation is a significant predictor of emotional intelligence of education students. This is aligned with the research of Shabani, et al. (2021), since the mothers who were barangay officials had very high emotional intelligence, who stated that the women's emotional intelligence was the key factor to effective leadership since they were emotional and emphatic. It can be implied that the mothers who had leadership positions were more emotionally intelligent compared to those who do not since the decisions that they made as leaders were critical and must be thought of thoroughly. With this, one may delve further into the effect of emotional intelligence of the women leaders into their companies or communities.

The students who have fathers who had no occupation ($M=3.99$, $SD=0.45$), other occupation ($M= 3.96$, $SD=0.35$), government employee ($M=3.81$, $SD=0.70$), and businessman/ entrepreneur have above average emotional intelligence. Those with fathers who were barangay officials ($M=3.15$) had average emotional intelligence. F-test revealed that there is a significant difference, $F(4, 79) = 2.682$, $p=.037$, on emotional intelligence when education students were grouped according to father's occupation. These findings suggest that father's occupation is a significant predictor of emotional intelligence of students of education students. This result contradicted Jyoti (2022) who found that father's occupation does not significantly predict the emotional intelligence of studnets, but agrees with Kusha (2017) who found that father's occupation significantly influences the emotional maturity of adolescents, impacting their emotional intelligence. Higher service class fathers correlate with higher emotional and social maturity students.

The students from a family earning P11, 000 – P15, 999 monthly had very high emotional intelligence. The rest of the students with P5, 999 and below ($M=3.91$, $SD=0.42$), P21,000 and above ($M=3.91$, $SD=0.60$), P16,000-P20,999 ($M=3.88$, $SD=0.00$), P6,000 – P10,999 ($M=3.87$, $SD=0.40$) as estimated monthly family income, had above average emotional intelligence. There is no significant difference, $F(4, 79) = .979$, $p=.424$, on emotional intelligence when education students were grouped according to estimated family income. These findings suggest that the income of the family where the students belong is not a significant predictor of emotional intelligence. This contrasts with the study of PitchaiahPodila (2018) which stated that male students from low-income families had higher emotional intelligence than those from high-income families and the females had higher emotional intelligence among the group. Based on this, it can be implied that those in medium-income families had higher

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 9 of 12

emotional intelligence because they had to struggle to climb the financial ladder and were also motivated not to fail to go lower in their social status.

Table 4

Level of Emotional Intelligence of Students when Grouped According to Profile

Profile	M	Emotional Intelligence SD	Interpretation	F-test/ t-test	p-value
Age					
17-19 years old	3.92	0.40	Above Average	.072	.975 ^{ns}
20-21 years old	3.92	0.45	Above Average		
22-23 years old	3.98	0.45	Above Average		
24 years old and above	3.97	0.11	Above Average		
Civil Status					
Single	3.93	0.41	Above Average	1.380	.171 ^{ns}
Married	4.34	0.47	Very High		
Mother's Occupation					
Teacher/Instructor	3.91	0.60	Above Average	1.635	.174 ^{ns}
Business Woman/ Entrepreneur	2.97	.	Above Average		
Barangay Official	4.25	0.52	Very High		
Government Employee	3.76	0.05	Above Average		
None	3.99	0.43	Above Average		
Others	3.88	0.36	Above Average		
Father's Occupation					
Businessman/Entrepreneur	2.97	.	Above Average	1.903	.103 ^{ns}
Barangay Official	3.15	.	Average		
Government Employee	3.81	0.70	Above Average		
None	3.99	0.45	Above Average		
Others	3.96	0.35	Above Average		
Estimated Monthly Family Income					
P5,999 and below	3.94	0.42	Above Average	.979	.424 ^{ns}
P6,000-P10,999	3.87	0.40	Above Average		
P11,000-15,999	4.73	.	Very High		
P16,000-P20,999	3.88	0.00	Above Average		
P21,000 and above	3.91	0.60	Above Average		
Grand Mean	3.94	0.42	Above Average		

Scale: 4.21-5.00 Very High Emotional Intelligence; 3.41-4.20 Above Average Emotional Intelligence; 2.61-3.40 Average Emotional Intelligence; 1.81-2.60 Below Average Emotional Intelligence; 1.00-1.80 Low Emotional Intelligence

**p-value is significant at 0.01 level; *p-value is significant at 0.05 level; ^{ns} p-value is non-significant at 0.05

Relationship between Grit and Emotional Intelligence of Students

A Pearson *r* was performed to determine the relationship between students' grit and emotional intelligence as shown in Table 5. A weak, positive correlation between grit and emotional intelligence was statistically significant ($r = .339, p = .002$). Furthermore, the results suggest that this observed correlation is statistically significant. This means it is unlikely to be due to chance and provides evidence of a real association between the two variables. This result is congruence to the study of Özer (2021) where the findings showed that there was a significant and constructive correlation between grit and emotional intelligence. The result also agreed to the study of Anto et al. (2023) which stated that there was a considerable positive correlation between emotional intelligence and decision- making as well as between grit and emotional intelligence. Based on these, it can be implied that those with high grit will also have high emotional intelligence. In pursuing their long-term goals, the respondents must also be emotionally intelligent and prepared, especially when drawbacks happen.

Table 5

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 10 of 12

Relationship between grit and emotional intelligence of students

Variables	Pearson r	Interpretation
Grit	$r = .339^{**}$	Weak significant correlation
Emotional Intelligence	$r = .002$	

Scale: .80 to 1.00 Very strong correlation; .60 to .799 Strong correlation; .40 to .599 moderate correlation; .20 to .399 Weak correlation; .000 to .199 Very weak correlation

$^{**}p < 0.05$; significant at 0.01

Relationship between Grit and Academic Resilience of Students

A Pearson r was performed to determine the relationship between students' grit and academic resilience as shown in Table 6. There was a weak, positive correlation between grit and academic resilience which was statistically significant ($p = .350$, $p = .001$). Furthermore, the results suggest that this observed correlation is statistically significant. This means it is unlikely to be due to chance and provides evidence of a real association between the two variables. This agreed with the study of Saifullah and Khan (2022) that grit and academic resilience had a positive and significant correlation. From this, it can be implied that as grit increases, academic resilience tends to increase also.

Table 6

Relationship between grit and academic resilience of students

Variables	Pearson r	Interpretation
Grit	$r = .350^{**}$	Weak significant correlation
Academic Resilience	$r = .001$	

Scale: .80 to 1.00 Very strong correlation; .60 to .799 Strong correlation; .40 to .599 moderate correlation; .20 to .399 Weak correlation; .000 to .199 Very weak correlation

$^{**}p < 0.05$; significant at 0.01

Relationship between Academic Resilience and Emotional Intelligence of Students

A Pearson r man's rank-order correlation was performed to determine the relationship between students' academic resilience and emotional intelligence as shown in Table 6. There was a strong, positive correlation between grit and emotional intelligence, which was statistically significant ($r = .694$, $p = .000$). This means that as academic resilience scores increase, emotional intelligence scores also tend to increase significantly. There is a clear association between the two variable. Students with higher academic resilience also tend to score significantly higher in emotional intelligence. This is in conjunction with the study of Ononye, et. al. (2022) who stated that academic resilience was positively related to emotional intelligence. It can be implied that as the grit increases, academic resilience tends to increase also.

Table 7

Relationship between academic resilience and emotional intelligence of students

Variables	Pearson r	Interpretation
Grit	$r = .694^{**}$	Strong significant correlation
Academic Resilience	$p = .000$	

Scale: .80 to 1.00 Very strong correlation; .60 to .799 Strong correlation; .40 to .599 moderate correlation; .20 to .399 Weak correlation; .000 to .199 Very weak correlation

$^{**}p < 0.05$; significant at 0.01

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 11 of 12

The results showed that the respondents were gritty, highly academic resilient, with above-average emotional intelligence. It further revealed there was a weak significant correlation among grit and emotional intelligence, and grit and academic resilience; and academic resilience and emotional intelligence had a strong significant correlation. Based on these results, it can be concluded that education students showed perseverance to attain their long- term goals and that they were determined to achieve these goals even if there were setbacks or problems that may hinder to achieve these goals immediately. It can also be concluded the respondents were able to surpass the challenges of the teacher education program and the pressure that comes along with it especially that it is a board program. With the results showing above average emotional intelligence, it can be concluded that the respondents were able to handle both positive and negative emotions well to be able to have a clear direction in life and be able to create sound decisions.

Based on these, it is recommended to create grit-development program as to help them pursue and be motivated to work towards their goals despite the different challenges they will encounter along the way. It is also recommended to integrate resilience-building strategies to help the students cope with stress, especially academically. It is also recommended to create a supportive learning environment so that the students will not be afraid to take risks, or that they will not be afraid of failure to support their emotional needs. It is further recommended that the institution promote well- being and mental of the students , especially creating balance between the academics and personal lives of the students, to for them to have stronger emotional intelligence.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

First and foremost, the researchers would like to thank the Almighty God for His guidance and provision in the completion of this study. The researchers would also like to thank to Jennifer P. Benlior, DPA, Satellite College Director, for her support Dr. Joesan Borres, Research Chair, for her encouragement in the completion of this study. Heartfelt thanks is also extended to the students of CAPSU Pilar for their cooperation as they served as respondents of this study; and to the family of the researchers, greatest gratitude is extended.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

- Alejandra, M., Osorno-Foronda., M., De, Hoyos-Rivera., M., García-Vallejo, A., V., Cordoba-Sanchez, V. (2021). El estrés académico y resiliencia en madres universitarias colombianas con y sin apoyo institucional en el cuidado de sus hijos. doi: 10.25057/21452776.1367
- Anto, I.T., Sebastian, A., Anil, A., & Mathews, K.D. (2023). The Relationship between Grit, Emotional Intelligence and Decision making among Emerging Adults. international journal of engineering technology and management sciences.
- Arco-Tirado, J. L., Bojica, A., Fernández-Martín, F., & Hoyle, R. H. (2019). Grit as Predictor of Entrepreneurship and Self-Employment in Spain. *Frontiers in Psychology*, 10. <https://doi.org/10.3389/fpsyg.2019.00389>
- Blair, C., Carstone., Juliana, Bell., Bryce, Carstone, Smith. (2024). Fostering academic resilience in higher education. *Journal of Perspectives in Applied Academic Practice*, doi: 10.56433/jpaap.v12i1.598
- Dibya, Mukherjee., Soumyajit, Roy. (2023). Cross-sectional study. doi: 10.1016/b978-0-323-88423-5.00101-1
- Jyoti, Kumari. (2022). Impact of parental occupation on emotional intelligence and moral values of secondary school students. *Scholarly research journal for humanity science & English language*, doi: 10.21922/srjhsel.v10i54.11787
- Kannangara, C. S., Allen, R. E., Waugh, G., Nahar, N., Khan, S. Z. N., Rogerson, S., & Carson, J. (2018). All That Glitters Is Not Grit: Three Studies of Grit in University Students. *Frontiers in Psychology*, 9. <https://doi.org/10.3389/fpsyg.2018.01539>
- Khodarahimi, S. (2015). The Role of Marital Status in Emotional Intelligence, Happiness, Optimism and Hope. *Journal of Comparative Family Studies*, 46(3), 351–371. <https://doi.org/10.3138/jcfs.46.3.351>

	Document Type:	DOCUMENTED INFORMATION		Document Code	RES- CR-01
		ISO 9001:2015		Revision No.	00
	Document Title:	COMPLETED RESEARCH		Effective Date	
				Page	Page 12 of 12

- Kundu, D. (2019). Is age a good predictor of emotional intelligence – a case study among private sector employees.
- Kusha, Pant., Ritu, Singh. (2017). Role of fathers in social and emotional development of their adolescents across their working profile. *Asian Journal of Home Science*, doi: 10.15740/HAS/AJHS/12.1/34-41
- Ononye, U., Ogbeta, M., Ndudi, F., Bereprebofa, D., & Maduemezia, I. (2022). Academic resilience, emotional intelligence, and academic performance among undergraduate students. *Knowledge and Performance Management*, 6(1), 1–10. [https://doi.org/10.21511/kpm.06\(1\).2022.01](https://doi.org/10.21511/kpm.06(1).2022.01)
- Özer, E. (2021). The Relationship between Grit and Emotional Intelligence in University Students. *Psycho-Educational Research Reviews*, 25–33. <https://www.journals.lapub.co.uk/index.php/perr/article/view/1708>
- PitchaiahPodila, S. (2018). EMOTIONAL INTELLIGENCE AND INCOME- A CASE STUDY. *International Journal of Advanced Research*, 6(12), 282–288. <https://doi.org/10.21474/ijar01/8141>
- Polit DF, Beck CT, Owen SV. Is the CVI an acceptable indicator of content validity? Appraisal and recommendations. *Res Nurs Health*. 2007 Aug;30(4):459-67. doi: 10.1002/nur.20199. PMID: 17654487.
- Puspa, Zuleika., Legiran. (2022). Cross-Sectional Study as Research Design in Medicine. *Archives of The Medicine and Case Reports*, doi: 10.37275/amcr.v3i2.193
- Robbins, A., Kaye, E., & Catling, J. (2018). Predictors of Student Resilience in Higher Education.. *Psychology Teaching Review*.
- Roxas, E. D. (n.d.). Level of Grit Among Faculty Members in a Selected Higher Academic Institution. In *The Southeast Asian Conference on Education 2020*.
- Saifullah, S., & Khan, O. Z. (2022). Relationship Between Grit and Academic Resilience Among University Students. *CARC Research in Social Sciences*, 1(1), 11–14. <https://doi.org/10.58329/criss.v1i1.4>
- Shabani, S., Citaku, F. T., Ramadani, H., Mano, M. S., Barach, P., Waldrop, M., Zillioux, D., & Khan, Y. H. (2021). The Impact of Emotional Intelligence on Enhancing Women's Leadership Competencies. *International Journal of Business and Management*, 16(11), 105. <https://doi.org/10.5539/ijbm.v16n11p105>
- Simones Landry, M. (2018). Exploring the Impact of Grit and Sponsorship on Female Leadership and Aspirations. *All Student Scholarship*. <https://digitalcommons.usm.maine.edu/etd/318/>
- Yan, Yan., Xiaosong, Gai. (2022). High Achievers from Low Family Socioeconomic Status Families: Protective Factors for Academically Resilient Students. *International Journal of Environmental Research and Public Health*, doi: 10.3390/ijerph192315882
- Ye, W., Strietholt, R. & Blömeke, S. Academic resilience: underlying norms and validity of definitions. *Educ Asse Eval Acc* **33**, 169–202 (2021). <https://doi.org/10.1007/s11092-020-09351-7>